

Industrial Internship Report on Smart City Traffic Forecasting

Prepared by

Shruti Vishnu Ma

Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "Smart City Traffic Forecasting." The objective of this project was to develop a machine learning-based traffic forecasting system capable of predicting future traffic volume using historical traffic data collected from multiple city junctions. The project involved several stages of the data science lifecycle, including data understanding, data preprocessing, feature engineering, exploratory data analysis (EDA), and predictive modeling to identify traffic patterns and forecast vehicle flow. The ultimate goal of the project was to support intelligent transportation systems by providing accurate traffic predictions that can assist city authorities in optimizing traffic management, reducing congestion, improving route planning, and enhancing urban mobility through data-driven decision-making.

Throughout the internship, I gained practical experience in applying machine learning concepts to solve a real-world problem. I also developed skills in data analysis, feature engineering, visualization, project documentation, and version control using Git and GitHub. Working on an industry-oriented project strengthened my problem-solving abilities and provided valuable insight into how data science projects are developed and executed in a professional environment.

This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

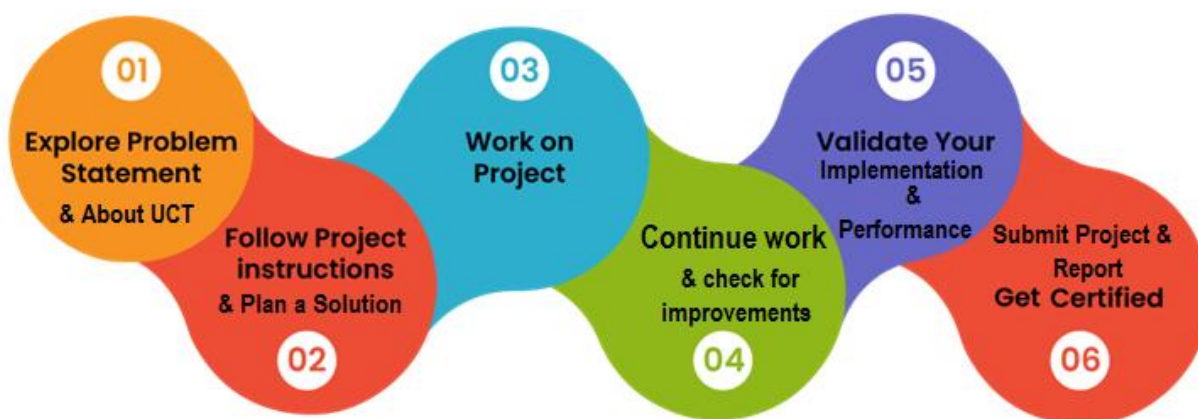
TABLE OF CONTENTS

1	Preface	3
2	Introduction	4
2.1	About UniConverge Technologies Pvt Ltd	4
2.2	About upskill Campus	8
2.3	Objective	10
2.4	Reference	10
2.5	Glossary	10
3	Problem Statement	11
4	Existing and Proposed solution	12
5	Proposed Design/ Model	13
5.1	High Level Diagram (if applicable)	14
5.2	Low Level Diagram (if applicable)	15
5.3	Interfaces (if applicable)	15
6	Performance Test	16
6.1	Test Plan/ Test Cases	17
6.2	Test Procedure	18
6.3	Performance Outcome	18
7	My learnings	21
8	Future work scope	22

1 Preface

The six-week industrial internship conducted by Upskill Campus (USC), The IoT Academy, and UniConverge Technologies Pvt. Ltd. (UCT) provided valuable industry exposure and enhanced my technical and problem-solving skills.

During the internship, I worked on the Smart City Traffic Forecasting project, gaining practical experience in data preprocessing, feature engineering, exploratory data analysis, and machine learning. Overall, this internship bridged the gap between academic learning and real-world applications, boosting my confidence and preparing me for a career in Data Science and Machine Learning.



During this internship, I gained practical experience in Machine Learning, data preprocessing, feature engineering, exploratory data analysis (EDA), and traffic forecasting. I also improved my problem-solving, analytical thinking, and project development skills while learning to use Python, Jupyter Notebook, Git, and GitHub effectively. Overall, this internship provided valuable industry exposure and enhanced my technical knowledge and confidence in working on real-world data science projects.

I would like to thank my mentors, faculty members, family, friends, Upskill Campus, The IoT Academy, and UniConverge Technologies Pvt. Ltd. (UCT) for their constant guidance and support throughout my internship.

During this internship, I gained practical experience in Machine Learning, data preprocessing, feature engineering, exploratory data analysis (EDA), and traffic forecasting. I also improved my problem-solving, analytical thinking, and project development skills while learning to use Python, Jupyter Notebook, Git, and GitHub effectively. Overall, this internship provided valuable industry exposure and enhanced my technical knowledge and confidence in working on real-world data science projects.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



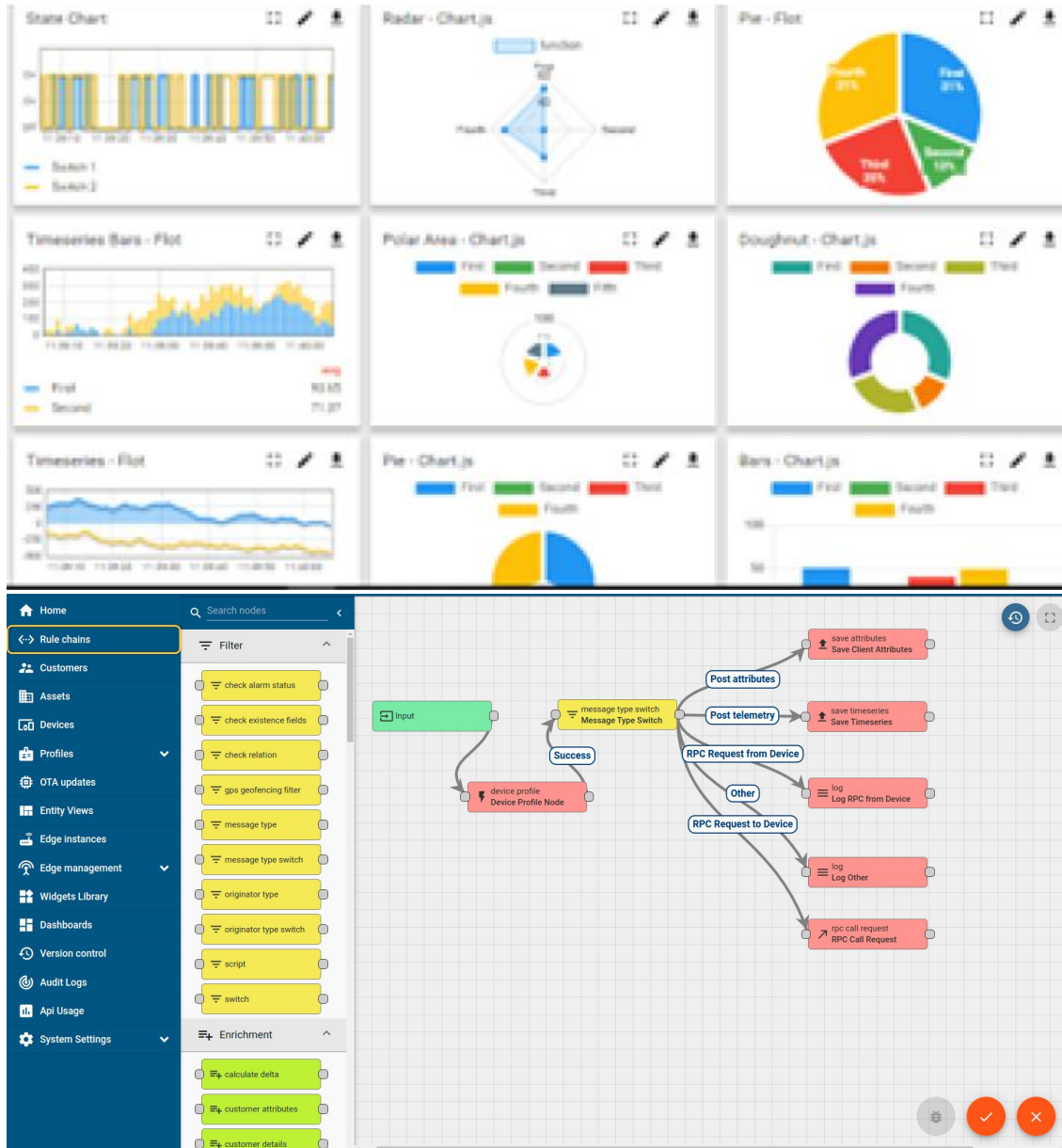
i. UCT IoT Platform (**Insight**)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



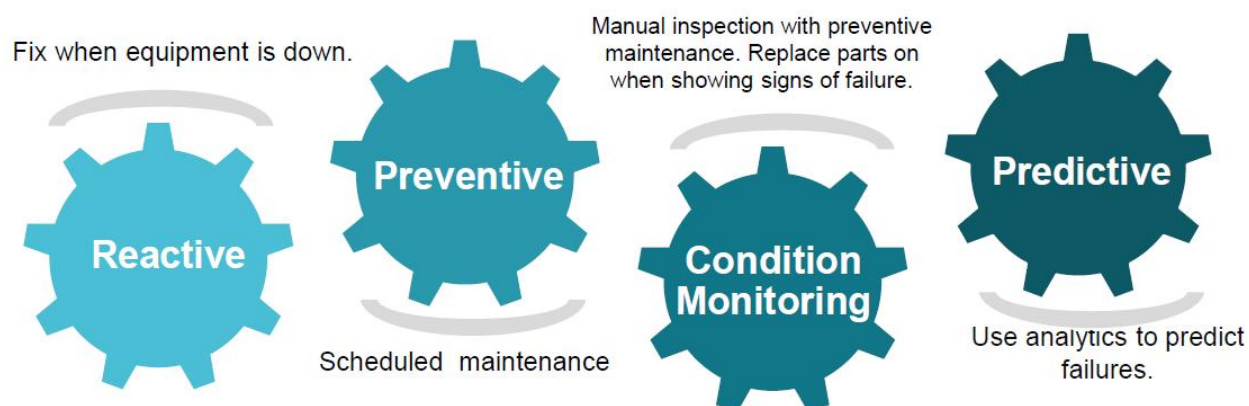


iii. based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

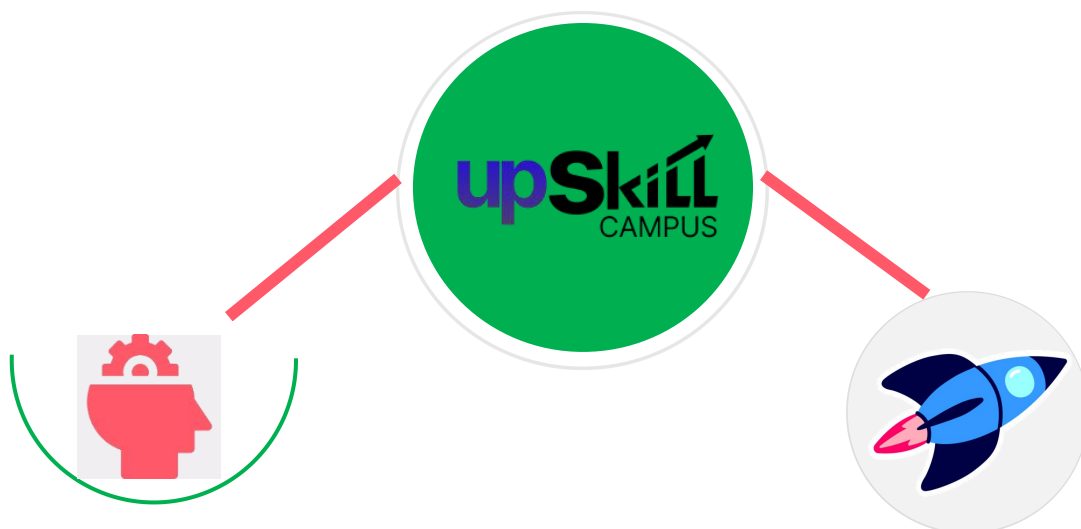
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

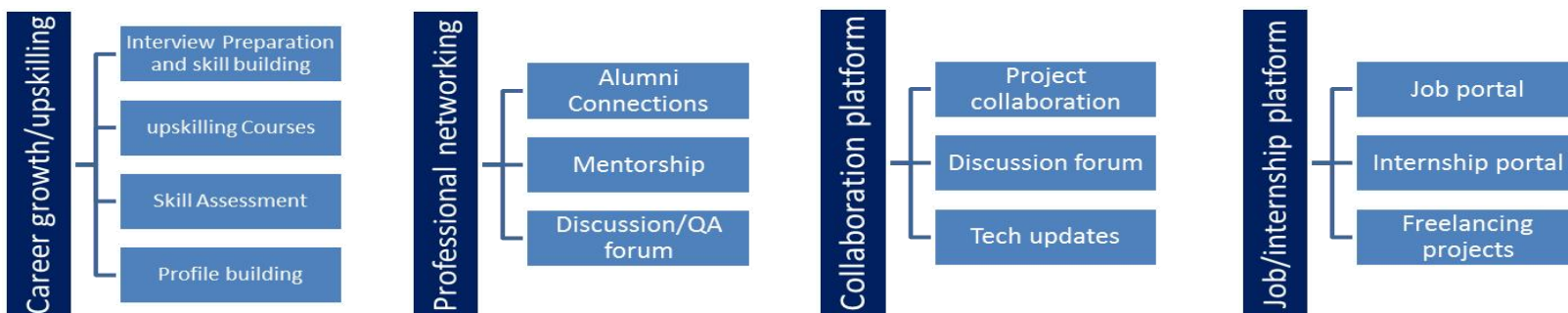
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Li, Gao. "Application of Data Twinning Based on Deep Time Series Model in Smart City Traffic Flow Prediction." Discover Internet of Things, Springer Nature, 2025
- [2] Ali, Ahmad, Amin Sharafian, H.M. Yasir Naeem, Muhammad Zakarya, Zongze Wu, and Xiaoshan Bai. "Advanced Computational Models for Urban Traffic Flow Prediction: A Comprehensive Review and Future Directions." Computer Science Review, Elsevier, 2026.
- [3] Qin, Z., Wang, M., Zhu, H., et al. "A Survey on Modern Deep Learning Techniques for Traffic Volume Prediction." Data Science and Engineering, Springer Nature, 2026.

2.6 Glossary

Terms	Acronym
Artificial Intelligence	AI
Machine Learning	ML
Data Science	DS
Exploratory Data Analysis	EDA
Internet of Things	IOT
Smart City	SC
Traffic Flow Prediction	TFP
Time Series Analysis	TSA
Long Short-Term Memory	LSTM
Random Forest	RF
Mean Absolute Error	MAE
Root Mean Squared Error	RMSE

3 Problem Statement

In the assigned problem statement

With rapid urbanization, population growth, and an increasing number of vehicles on roads, traffic congestion has become a major challenge in modern cities. Traditional traffic management systems mostly depend on current traffic conditions and manual monitoring, making it difficult to anticipate future traffic patterns. This creates a need for intelligent systems that can analyze historical data and predict upcoming traffic conditions.

The objective of this project is to develop a Smart City Traffic Forecasting system that uses historical traffic data to predict future vehicle volume at different road junctions. By analyzing past traffic records, the system identifies patterns related to time, day, and junction-wise traffic movement, helping in understanding traffic behavior and improving decision-making.

The project applies Data Science and Machine Learning techniques such as data preprocessing, exploratory data analysis, feature engineering, and predictive modeling to build an effective forecasting model. Time-based features such as hour, day, month, and weekday are extracted from the dataset to improve the accuracy of traffic predictions.

The proposed system can assist smart city authorities in better traffic management by providing future traffic insights. Accurate traffic forecasting can help reduce congestion, optimize traffic signal control, improve route planning, and support the development of efficient intelligent transportation systems.

4 Existing and Proposed solution

Existing Solutions

Several existing traffic forecasting systems use statistical methods, machine learning algorithms, and sensor-based monitoring to estimate traffic flow. Modern navigation applications and intelligent transportation systems provide real-time traffic updates using GPS data, IoT sensors, and historical traffic information. While these solutions are effective for monitoring current traffic conditions, many require expensive infrastructure, continuous real-time data collection, and significant computational resources.

Limitations of Existing Solutions

Most existing solutions focus on real-time traffic monitoring rather than accurate long-term traffic forecasting. They often depend on large-scale IoT deployments, GPS tracking, or cloud-based infrastructure, making them costly to implement and maintain. Some models also struggle to capture changing traffic patterns caused by holidays, weather conditions, or unexpected events, leading to reduced prediction accuracy.

Proposed Solution

The proposed solution is a Machine Learning-based Smart City Traffic Forecasting System that predicts future traffic volume using historical traffic data. The system performs data preprocessing, exploratory data analysis, feature engineering, and predictive modeling to identify traffic patterns at different city junctions. By learning from past traffic trends, the model generates forecasts that can assist authorities in planning and managing urban traffic more efficiently.

Value Addition

The proposed system offers a cost-effective and data-driven approach to traffic prediction without requiring complex real-time sensor networks. It helps improve traffic management by providing advance traffic forecasts that can support route planning, congestion reduction, and better traffic signal management. The project also demonstrates the practical application of Data Science and Machine Learning techniques, making it scalable and suitable for future integration with smart city infrastructure, real-time IoT data, and advanced deep learning models.

4.1 Code submission (Github link)

<https://github.com/Shrutimane3007/upskillcampus.git>

<https://github.com/Shrutimane3007/upskillcampus/blob/main/app.py>

4.2 Report submission (Github link) :

https://github.com/Shrutimane3007/upskillcampus/blob/main/SmartCityTrafficForecasting_Shruti_USC_UCT.pdf

5 Proposed Design/ Model

The Smart City Traffic Forecasting System is designed to predict future traffic volume using historical traffic data and Machine Learning techniques. The application begins by loading the traffic dataset and the pre-trained Random Forest Regression model. Before making predictions, the dataset undergoes data preprocessing, where missing values and inconsistencies are handled, and the DateTime attribute is converted into a proper date-time format. This ensures that the data is clean, structured, and suitable for further analysis and model prediction.

After preprocessing, the system performs feature engineering by extracting important temporal features such as Year, Month, Day, Hour, Day of Week, Weekend, Quarter, and Week of Year from the DateTime column. These features help the model understand daily, weekly, and seasonal traffic patterns, allowing it to learn complex relationships between time and vehicle movement. The processed features are then provided as input to the trained Random Forest Regression model, which generates accurate traffic volume predictions based on historical trends.

The Streamlit-based web application provides three major modules to improve user interaction. The Dashboard module presents key traffic statistics and interactive visualizations, including hourly traffic trends, monthly traffic analysis, and junction-wise vehicle distribution. The Traffic Prediction module allows users to enter details such as junction, date, and time to obtain traffic predictions for a specific instance. The Batch Prediction module enables users to upload a CSV file containing multiple traffic records, allowing the system to predict traffic volume for all records simultaneously.

Finally, the system displays the predicted vehicle count and classifies the traffic condition into Low, Moderate, or Heavy based on the prediction results. The application also provides graphical visualizations of predicted traffic patterns and offers the option to download the prediction results as a CSV file. This complete workflow enables efficient traffic analysis, supports informed decision-making for traffic authorities, and contributes to better traffic management, route optimization, and future smart city transportation planning.

5.1 High Level Diagram (if applicable)

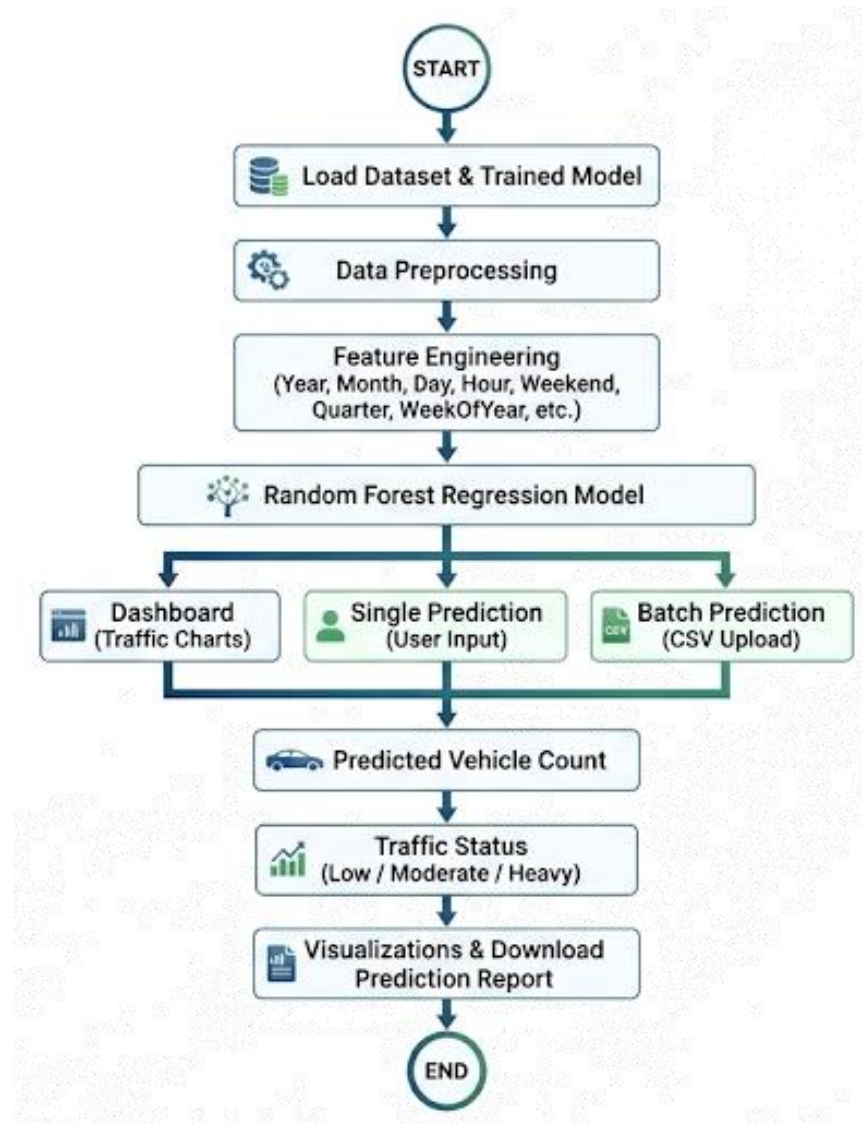
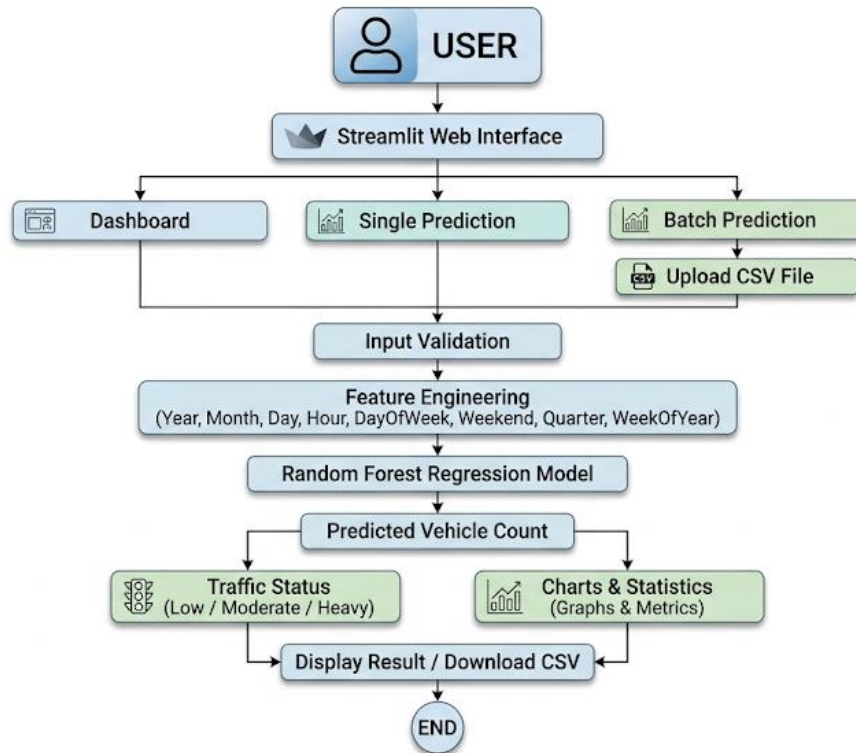


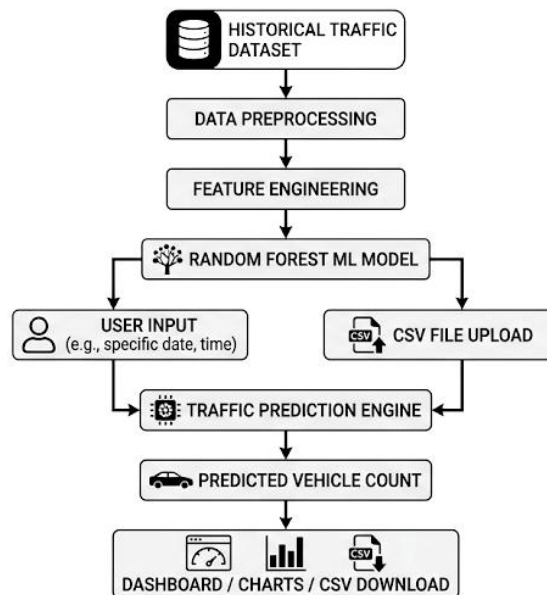
FIGURE 1: HIGH LEVEL DIAGRAM OF THE PROPOSED SMART CITY TRAFFIC FORECASTING SYSTEM

5.2 Low Level Diagram (if applicable)



5.3 Interfaces (if applicable)

TRAFFIC VOLUME PREDICTION DFD SYSTEM PIPELINE



6 Performance Test

The performance of the Smart City Traffic Forecasting system was evaluated based on several practical constraints that are important for real-world traffic prediction applications. These include prediction accuracy, response time, memory usage, scalability, and reliability. The proposed system uses a Random Forest Regression model, which provides a good balance between prediction accuracy and execution speed while efficiently handling historical traffic data.

Constraint	Impact on System	Solution Adopted	Result
Prediction Accuracy	Incorrect predictions may affect traffic planning.	Used feature engineering and Random Forest Regression.	Achieved reliable traffic predictions on the test dataset.
Processing Speed	Slow prediction reduces user experience.	Pre-trained model used for prediction instead of retraining each time.	Predictions generated within a few seconds.
Memory Usage	Large datasets require higher memory.	Loaded only the required dataset and trained model into memory.	Suitable for execution on a standard personal computer.
Scalability	Increasing traffic data may increase processing time.	Batch prediction feature supports prediction for multiple records.	Can process larger datasets with acceptable performance.
Reliability	Invalid user inputs may cause errors.	Input validation and exception handling implemented.	Stable execution with proper error messages.

The application was tested using historical traffic data consisting of approximately 48,120 records. The trained Random Forest Regression model successfully generated traffic predictions for both individual user inputs and batch CSV files. The dashboard displayed traffic statistics and interactive visualizations without significant delay. The system remained responsive during prediction and data visualization tasks on a standard laptop.

Although the developed system performs efficiently for the available dataset, several factors may affect its performance in real-world deployment.

- Very large datasets may increase memory usage and prediction time.
- Traffic conditions influenced by accidents, road closures, weather, or public events are not included in the current model.
- The current model relies on historical traffic data and does not perform real-time traffic prediction.
- Continuous model retraining may be required as new traffic patterns emerge over time.

To overcome these limitations, the system can be enhanced by integrating real-time IoT sensor data, weather information, GPS-based traffic feeds, and advanced deep learning models such as LSTM or GRU. Deploying the model on cloud platforms with automated retraining can further improve scalability and prediction accuracy.

6.1 Test Plan/ Test Cases

Test Case ID	Test Scenario	Test Input	Expected Result	Actual Result	Status
TC-01	Load Application	Launch Streamlit application	Application loads successfully with dashboard	Application loaded successfully	Pass
TC-02	Load Dataset	Valid training dataset	Dataset loads without errors	Dataset loaded successfully	Pass
TC-03	Load ML Model	Valid model.pkl file	Random Forest model loads successfully	Model loaded successfully	Pass
TC-04	Single Traffic Prediction	Valid Junction, Date and Time	Predicted vehicle count is displayed	Prediction generated correctly	Pass
TC-05	Invalid Date Input	Invalid calendar date	Error message displayed	Appropriate error displayed	Pass
TC-06	Batch Prediction	Upload valid CSV file	Predictions generated for all records	Batch prediction successful	Pass
TC-07	Invalid CSV Format	CSV missing required columns	Validation error displayed	Error message displayed	Pass
TC-08	Dashboard Visualization	Open Dashboard	Charts and traffic statistics displayed	Dashboard displayed successfully	Pass
TC-09	Download Prediction	Click Download CSV	Prediction results downloaded	CSV downloaded successfully	Pass
TC-10	Traffic Classification	Predicted traffic value	Display Low/Moderate/Heavy traffic status	Correct traffic status displayed	Pass

The Smart City Traffic Forecasting system was tested to verify its functionality, prediction capability, and user interaction. Testing was carried out in a systematic manner to ensure that each module of the application performed as expected under different input conditions. Initially, the Streamlit application was launched and verified for successful loading of the historical traffic dataset and the pre-trained Random Forest Regression model. The Dashboard module was tested to ensure that traffic statistics, graphs, and visualizations were displayed correctly without any errors.

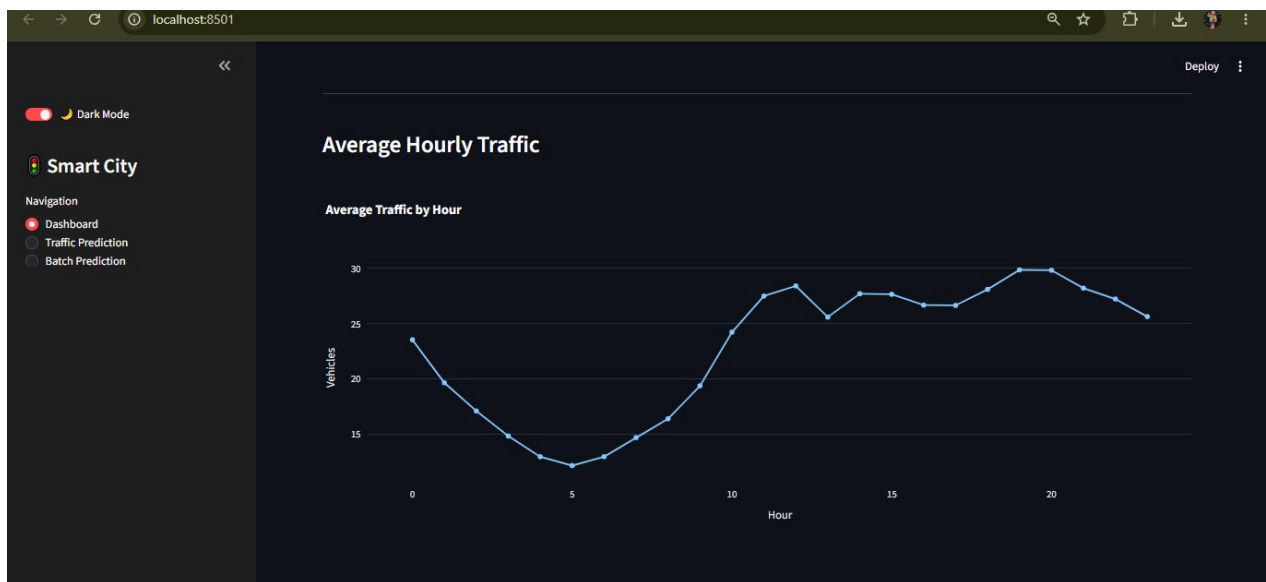
Next, the Single Traffic Prediction module was tested by providing different combinations of junction numbers, dates, and time values. The predicted vehicle count and corresponding traffic status (Low, Moderate, or Heavy) were verified for correctness. Invalid inputs, such as incorrect dates, were also tested to ensure that the application displayed appropriate error messages and handled exceptions gracefully.

The Batch Prediction module was evaluated by uploading CSV files containing multiple traffic records. The system successfully processed the uploaded dataset, generated predictions for all records, displayed the results in tabular form, and allowed users to download the prediction output as a CSV file. The application was also tested for responsiveness, stability, and proper navigation between modules. Throughout the testing process, all major functionalities operated successfully, confirming that the system performs reliably for traffic forecasting and analysis.

6.3 Performance Outcome

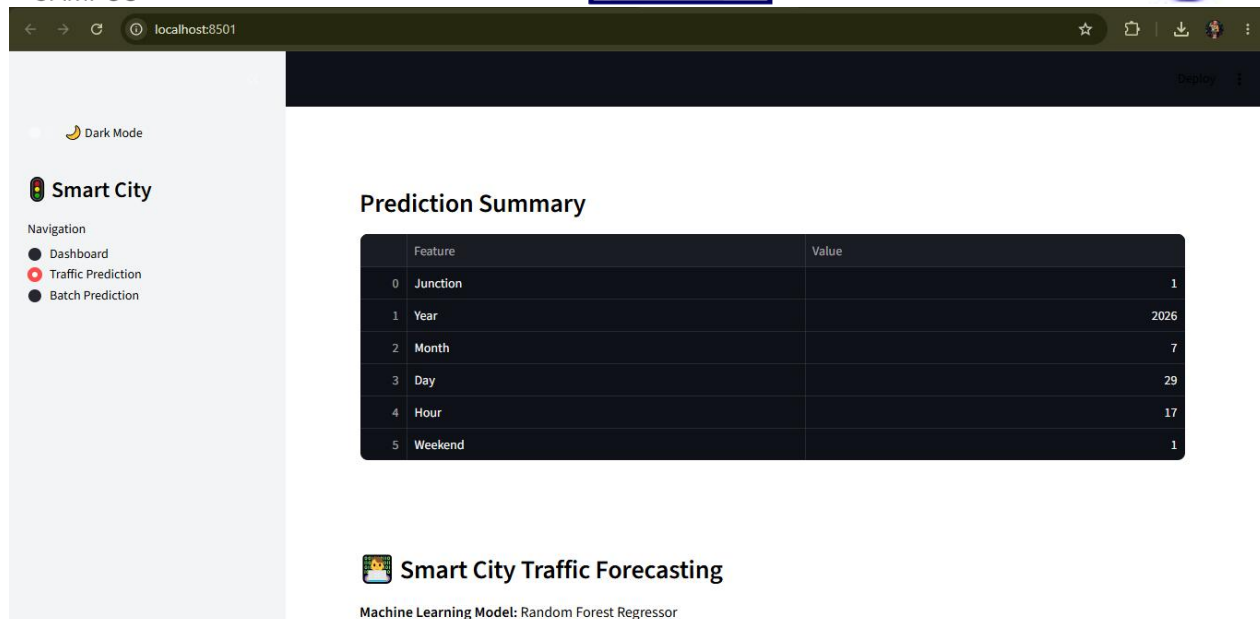
The Smart City Traffic Forecasting system successfully predicted traffic volume using the Random Forest Regression model. The application generated accurate predictions for both single and batch inputs while maintaining smooth performance and a user-friendly interface. The Dashboard displayed traffic statistics and visualizations efficiently, and the prediction modules produced results within a short time. Input validation ensured reliable operation by handling invalid data appropriately.

Overall, the system met its objectives by providing accurate traffic forecasts, interactive visualizations, and downloadable prediction reports, making it suitable for traffic analysis and future smart city applications.





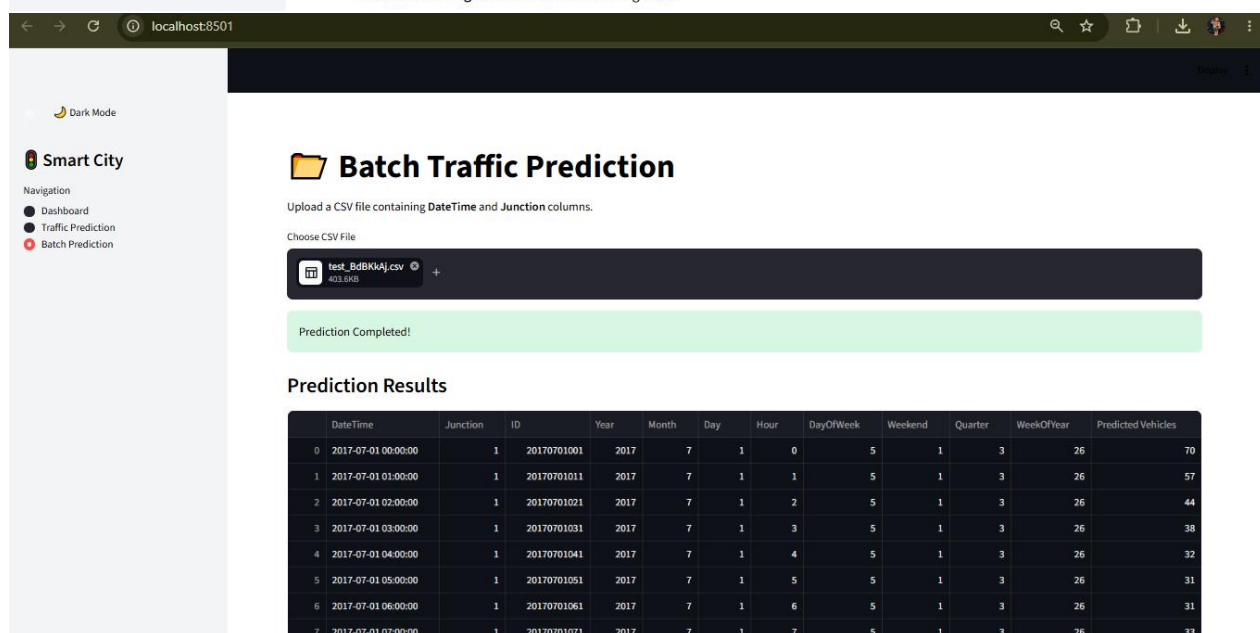
The screenshot shows a web application interface for a Smart City project. The left sidebar contains a 'Dark Mode' toggle and a 'Smart City' navigation menu with options: Dashboard, Traffic Prediction, and Batch Prediction. The main content area displays a 'Live Traffic Prediction' form. The form includes input fields for 'Select Junction' (1), 'Year' (2026), 'Month' (7), 'Day' (29), 'Hour' (17), and 'Day of Week' (Saturday). A 'Predict Traffic' button is present. Below the form, a green message box states 'Prediction Completed Successfully!'. A box displays 'Predicted Vehicles: 58 Vehicles'. At the bottom, a red status bar indicates 'Traffic Status: HEAVY'.



The screenshot shows a web browser at localhost:8501 displaying the 'Smart City' application. The left sidebar contains a 'Dark Mode' toggle and a 'Navigation' menu with options: Dashboard, Traffic Prediction (selected), and Batch Prediction. The main content area is titled 'Prediction Summary' and displays a table with the following data:

	Feature	Value
0	Junction	1
1	Year	2026
2	Month	7
3	Day	29
4	Hour	17
5	Weekend	1

Below the table, the title 'Smart City Traffic Forecasting' is shown, followed by the text 'Machine Learning Model: Random Forest Regressor'.



The screenshot shows the 'Batch Traffic Prediction' section of the Smart City application. It prompts the user to 'Upload a CSV file containing DateTime and Junction columns.' and provides a 'Choose CSV File' button. A file named 'test_BdKKAJ.csv' (403.5KB) is shown as uploaded. A green banner indicates 'Prediction Completed!'. Below this, the 'Prediction Results' are displayed in a table:

	DateTime	Junction	ID	Year	Month	Day	Hour	DayOfWeek	Weekend	Quarter	WeekOfYear	Predicted Vehicles
0	2017-07-01 00:00:00	1	20170701001	2017	7	1	0	5	1	3	26	70
1	2017-07-01 01:00:00	1	20170701011	2017	7	1	1	5	1	3	26	57
2	2017-07-01 02:00:00	1	20170701021	2017	7	1	2	5	1	3	26	44
3	2017-07-01 03:00:00	1	20170701031	2017	7	1	3	5	1	3	26	38
4	2017-07-01 04:00:00	1	20170701041	2017	7	1	4	5	1	3	26	32
5	2017-07-01 05:00:00	1	20170701051	2017	7	1	5	5	1	3	26	31
6	2017-07-01 06:00:00	1	20170701061	2017	7	1	6	5	1	3	26	31
7	2017-07-01 07:00:00	1	20170701071	2017	7	1	7	5	1	3	26	33

7 My learnings

The six-week industrial internship provided me with valuable practical exposure to Data Science, Machine Learning, and real-world software development. I learned the complete workflow of a data science project, including data collection, preprocessing, exploratory data analysis (EDA), feature engineering, model building, evaluation, and deployment using Streamlit. Working on the Smart City Traffic Forecasting project enhanced my understanding of handling real-world datasets and developing predictive models using the Random Forest Regression algorithm.

During the internship, I strengthened my technical skills in Python, Pandas, NumPy, Scikit-learn, Plotly, and Streamlit. I also improved my ability to analyze data, visualize trends, solve practical problems, and build an interactive web application for machine learning predictions. Additionally, I gained experience in project documentation, testing, debugging, and presenting technical solutions in a structured manner.

This internship has significantly contributed to my professional growth by improving my analytical thinking, problem-solving abilities, and software development skills. The knowledge and hands-on experience gained through this project have increased my confidence in working on real-world Data Science and Machine Learning applications. These skills will be valuable in my future career as a Data Scientist or Machine Learning Engineer, enabling me to contribute effectively to industry projects and continuously enhance my technical expertise.

8 Future work scope

Although the Smart City Traffic Forecasting system successfully predicts traffic volume using historical data, there are several enhancements that can be implemented in the future. Due to time constraints, these features were not included in the current project but can significantly improve its performance and real-world applicability.

Integrate real-time traffic data from IoT sensors, GPS devices, or traffic cameras for live traffic prediction. Include external factors such as weather conditions, road accidents, holidays, and public events to improve prediction accuracy.

Replace or compare the current Random Forest model with advanced deep learning models such as LSTM, GRU, or Transformer-based time-series forecasting models.

Deploy the application on a cloud platform and develop a mobile-friendly interface for easier access by traffic authorities and commuters.

Integrate the system with Google Maps APIs or other navigation services to provide intelligent route recommendations and real-time congestion alerts.

These enhancements would make the system more scalable, accurate, and suitable for deployment in real-world smart city transportation environments.